# E-COMMERCE SALES ANALYSIS

Project Report | Python • MySQL • Power BI

## 1. Project Overview

The E-Commerce Sales Analysis project focuses on analyzing sales and profit data to understand business performance, product and category contribution, monthly trends, payment preferences, and geographic performance.

**Objective:** Convert raw e-commerce transaction data into meaningful business insights using Python, SQL, and Power BI.

### Technologies Used

• Python, Pandas and NumPy – data cleaning and preprocessing
• MySQL – data storage and SQL querying
• SQLAlchemy – Python-to-MySQL connectivity
• Microsoft Power BI – interactive dashboards
• Microsoft Excel – source dataset

## 2. Project Objectives

• Analyze overall sales and profit performance.
• Identify top-performing products and categories.
• Understand monthly sales trends.
• Analyze profit across cities and states.
• Understand payment-mode performance.
• Build interactive dashboards to present business performance.

## 3. Dataset Description

The dataset contains order ID and date, customer details, city and state, product and category, quantity, unit price, discount, payment mode, order status, and delivery date.

## 4. Data Cleaning and Preparation

Data preparation was performed using Python. The workflow included checking missing values and duplicate records, removing duplicate records where appropriate, standardizing selected text fields, converting date columns to datetime, and converting quantity, unit price, and discount to numeric formats.

**Calculated fields**
Sales = Quantity × Unit Price
Net Amount = Sales – (Sales × Discount / 100)

*Note: The Net Amount calculation assumes Discount is stored as a percentage.*

# POWER BI DASHBOARD & BUSINESS INSIGHTS

## 5. Power BI Dashboard

### Page 1 – Sales Analysis

| Total Sales | Total Profit | Total Orders | Average Order Value |
|---|---|---|---|
| 3.17M | 633.16K | 299 | 10.59K |

Visuals: Sales by Month, Sales by Category, Sales by Product, and Sales by Payment Mode. Slicers: Month and Product.

### Page 2 – Profit Analysis

Visuals: Profit by Month, Profit by Payment Mode, Profit by City, Profit by Category, Profit by Product, and Profit by State. Slicers: City, Category, and Month.

## 6. Business Insights

**Insight 1 – Category Performance**
Electronics generated the highest sales at approximately 1,288K, followed by Fashion (871K), Furniture (667K), and Home (594K).

**Insight 2 – Monthly Sales Performance**
May recorded the highest sales at approximately 770.61K, followed by June at 705.51K. March recorded the lowest sales at approximately 269.48K.

**Insight 3 – City-wise Profit Performance**
Pune generated approximately 151K profit, followed by Bengaluru at 136K. Lucknow and Delhi generated approximately 99K each.

**Insight 4 – Product-wise Profit**
Mouse generated approximately 128K profit, Chair 123K, Laptop 116K, and Bottle 106K. These are among the leading products by profit in the dashboard.

**Insight 5 – State-wise Profit**
Karnataka generated approximately 228K profit, followed by Delhi at 136K. Maharashtra and Rajasthan generated approximately 99K each.

# PROJECT CONCLUSION

## 7. Data Quality Considerations

The cleaning output shared during the project showed missing values in Order_Date, Unit_Price, Delivery_Date, and dependent calculated fields such as Sales, Net_Amount, and Profit. Missing Order_Date values also affect derived Month, Year, and weekday fields.

### Important considerations

- Do not fill missing Unit_Price with an arbitrary value; investigate the source or document an appropriate exclusion rule.
- Do not replace missing Delivery_Date with Order_Date unless same-day delivery is confirmed.
- Confirm that Discount is a percentage before applying the Net Amount formula.
- Check whether negative quantities represent returns before removing them.
- Protect customer names, email addresses, and phone numbers when sharing the dataset.

## 8. Conclusion

The E-Commerce Sales Analysis project demonstrates an end-to-end data analytics workflow using Python, MySQL, and Power BI. The analysis highlights sales and profit patterns across products, categories, months, cities, and states. The dashboards provide a consolidated view of business performance and help communicate findings from transactional data.

Dashboard figures are reproduced approximately from the supplied dashboard PDF.